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Answer

Of course! Let's break down this problem. It's a fantastic question that tests your core understanding of what an equivalence relation truly is. You've marked option (3) 5, which is absolutely correct! Awesome job. Now, let's understand *why* it's 5 in a step-by-step manner. **### Problem Analysis** ****Topic:**** Sets, Relations, and Functions ****Problem Type:**** ****Conceptual**** ****Core Concept:**** The number of equivalence relations on a set is equal to the number of ways you can ****partition**** the set into non-empty, disjoint subsets. Each partition corresponds to a unique equivalence relation. Let the set be ****A = {1, 2, 3}****. --- **### Method 1: The Partition Method (The Smart & Fast Way)** Think of an equivalence relation as a way to group elements that are "equivalent" to each other. These groups are called ****equivalence classes****. The collection of all these groups (equivalence classes) forms a partition of the set. A ****partition**** of a set is a way of dividing its elements into non-empty, non-overlapping subsets that cover the entire set. Let's find all the possible ways to partition the set **A = {1, 2, 3}**. ****Case 1: All elements are in one single group.**** ****Partition:**** $\{\{1, 2, 3\}\}$ ****Visual:**** $[1\ 2\ 3]$ (Imagine all three elements in one box) ****Meaning:**** Every element is related to every other element. This is the largest possible

equivalence relation, also known as the universal relation. * **Number of ways:**

1 **Case 2: The elements are split into one group of 2 and one group of 1.** *

Partitions: 1. $\{\{1, 2\}, \{3\}\}$ (1 and 2 are related, 3 is on its own) 2. $\{\{1, 3\}, \{2\}\}$ (1 and 3 are related, 2 is on its own) 3. $\{\{2, 3\}, \{1\}\}$ (2 and 3 are related, 1 is on its own) * **Visual:** * $[1\ 2] [3]$ * $[1\ 3] [2]$ * $[2\ 3] [1]$ * **Number of

ways:** 3 (This is the same as choosing 2 elements to be in a group, ${}^3C_2 = 3$)

Case 3: All elements are in their own separate groups. * **Partition:** $\{\{1\},$

$\{2\}, \{3\}\}$ * **Visual:** $[1] [2] [3]$ (Each element in its own box) * **Meaning:**

Each element is only related to itself. This is the smallest possible equivalence relation (Reflexive only), also known as the identity relation. * **Number of

ways:** 1 **Total Count:** Summing up all the possibilities: 1 (from Case 1) + 3 (from Case 2) + 1 (from Case 3) = **5**. So, there are **5** possible partitions, which means there are **5** non-empty equivalence relations on the set $\{1, 2, 3\}$. --- ### Method 2: Building the Relations Manually (The Foundational Way)

This method is slower but helps in building intuition. An equivalence relation must be **Reflexive (R)**, **Symmetric (S)**, and **Transitive (T)**. For the set $A = \{1, 2, 3\}$, any equivalence relation **must** contain $\{(1,1), (2,2), (3,3)\}$ to be reflexive.

Let's call this the base set, R_0 . 1. **The Smallest Relation (Identity):** * $R_1 = \{(1,1), (2,2), (3,3)\}$ * This is R, S, and T. It corresponds to the partition $\{\{1\}, \{2\}, \{3\}\}$. (1 relation) 2. **Adding one symmetric pair to R_0 :** * $R_2 = \{(1,1), (2,2), (3,3), (1,2), (2,1)\}$ * This is R, S, and T. Corresponds to $\{\{1, 2\}, \{3\}\}$. * $R_3 = \{(1,1), (2,2), (3,3), (1,3), (3,1)\}$ * This is R, S, and T. Corresponds to $\{\{1, 3\}, \{2\}\}$. * $R_4 = \{(1,1), (2,2), (3,3), (2,3), (3,2)\}$ * This is R, S, and T. Corresponds to $\{\{2, 3\}, \{1\}\}$. (3 relations of this type) 3. **Adding all possible pairs (Universal Relation):** * $R_5 = \{(1,1), (2,2), (3,3), (1,2), (2,1), (1,3), (3,1), (2,3), (3,2)\}$ * This is R, S, and T.

Corresponds to the partition $\{\{1, 2, 3\}\}$. (1 relation) * **Note:** If you add two symmetric pairs, say $\{(1,2), (2,1)\}$ and $\{(2,3), (3,2)\}$, transitivity forces you to add $\{(1,3)\}$ and $\{(3,1)\}$, which leads to this universal relation. Again, the total count is $1 + 3 + 1 = \mathbf{5}$. --- ### Key Takeaway & Further Learning

* The number of ways to partition a set of n elements is given by the **n -th Bell Number**, denoted by B_n . * For $n=1$, $B_1 = 1$ $\{\{1\}\}$ * For $n=2$, $B_2 = 2$ $\{\{1,2\}\}$, $\{\{1\},\{2\}\}$ * For $n=3$, **$B_3 = 5$** (This was our question!) * For $n=4$, $B_4 = 15$ Remembering the first few Bell numbers can save you a lot of time in an exam! ### Practice Suggestions 1. **PYQ:** Look for JEE Main questions asking for the number of equivalence relations or the number of onto functions (which also relates to partitions via Stirling Numbers). This is a recurring concept. 2. **NCERT:** Re-read the "Types of Relations" section in the Class 12 NCERT Chapter 1 (Relations

and Functions) to solidify your understanding of Reflexive, Symmetric, and Transitive properties. 3. **Challenge Yourself:** Try to list all 15 partitions for the set $\{1, 2, 3, 4\}$. This will give you a great feel for how the combinations work. You nailed it! Keep connecting concepts like this, and you'll do great. Keep up the hard work

Of course! Let's break down this problem. It's a fantastic question that tests your core understanding of what an equivalence relation truly is.

You've marked option (3) 5, which is absolutely correct! Awesome job.

Now, let's understand *why* it's 5 in a step-by-step manner.

Problem Analysis

- **Topic:** Sets, Relations, and Functions
 - **Problem Type:** **Conceptual**
 - **Core Concept:** The number of equivalence relations on a set is equal to the number of ways you can **partition** the set into non-empty, disjoint subsets. Each partition corresponds to a unique equivalence relation.

Let the set be $A = \{1, 2, 3\}$.

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Method 1: The Partition Method (The Smart & Fast Way)

Think of an equivalence relation as a way to group elements that are "equivalent" to each other. These groups are called **equivalence classes**. The collection of all these groups (equivalence classes) forms a partition of the set.

A **partition** of a set is a way of dividing its elements into non-empty, non-overlapping subsets that cover the entire set.

Let's find all the possible ways to partition the set $A = \{1, 2, 3\}$.

Case 1: All elements are in one single group.

- **Partition:** $\{\{1, 2, 3\}\}$
- **Visual:** $[1\ 2\ 3]$ (Imagine all three elements in one box)
- **Meaning:** Every element is related to every other element. This is the largest possible equivalence relation, also known as the universal relation.
- **Number of ways:** 1

Case 2: The elements are split into one group of 2 and one group of 1.

- **Partitions:**

1. $\{\{1, 2\}, \{3\}\}$ (1 and 2 are related, 3 is on its own)
2. $\{\{1, 3\}, \{2\}\}$ (1 and 3 are related, 2 is on its own)
3. $\{\{2, 3\}, \{1\}\}$ (2 and 3 are related, 1 is on its own)

- **Visual:**

- $\begin{bmatrix} 1 & 2 \\ 3 \end{bmatrix}$
- $\begin{bmatrix} 1 & 3 \\ 2 \end{bmatrix}$
- $\begin{bmatrix} 2 & 3 \\ 1 \end{bmatrix}$

- **Number of ways:** 3 (This is the same as choosing 2 elements to be in a group, ${}^3C_2 = 3$)

Case 3: All elements are in their own separate groups.

- **Partition:** $\{\{1\}, \{2\}, \{3\}\}$

- **Visual:** $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ (Each element in its own box)

- **Meaning:** Each element is only related to itself. This is the smallest possible equivalence relation (Reflexive only), also known as the identity relation.

- **Number of ways:** 1

Total Count:

Summing up all the possibilities: 1 (from Case 1) + 3 (from Case 2) + 1 (from Case 3) = **5**.

So, there are **5** possible partitions, which means there are **5** non-empty equivalence relations on the set $\{1, 2, 3\}$.

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Method 2: Building the Relations Manually (The Foundational Way)

This method is slower but helps in building intuition. An equivalence relation must be **Reflexive (R)**, **Symmetric (S)**, and **Transitive (T)**.

For the set $A = \{1, 2, 3\}$, any equivalence relation **must** contain $\{(1,1), (2,2), (3,3)\}$ to be reflexive. Let's call this the base set, R_0 .

1. **The Smallest Relation (Identity):**

- $R_1 = \{(1,1), (2,2), (3,3)\}$

- This is R, S, and T. It corresponds to the partition $\{\{1\}, \{2\}, \{3\}\}$. (1 relation)

2. Adding one symmetric pair to R_0 :

- $R_2 = \{(1,1), (2,2), (3,3), (1,2), (2,1)\}$
- This is R, S, and T. Corresponds to $\{\{1, 2\}, \{3\}\}$.
 - $R_3 = \{(1,1), (2,2), (3,3), (1,3), (3,1)\}$
- This is R, S, and T. Corresponds to $\{\{1, 3\}, \{2\}\}$.
 - $R_4 = \{(1,1), (2,2), (3,3), (2,3), (3,2)\}$
- This is R, S, and T. Corresponds to $\{\{2, 3\}, \{1\}\}$. (3 relations of this type)

3. Adding all possible pairs (Universal Relation):

- $R_5 = \{(1,1), (2,2), (3,3), (1,2), (2,1), (1,3), (3,1), (2,3), (3,2)\}$
- This is R, S, and T. Corresponds to the partition $\{\{1, 2, 3\}\}$. (1 relation)
- **Note:** If you add two symmetric pairs, say $(1,2), (2,1)$ and $(2,3), (3,2)$, transitivity forces you to add $(1,3)$ and $(3,1)$, which leads to this universal relation.

Again, the total count is $1 + 3 + 1 = 5$.

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Key Takeaway & Further Learning

- The number of ways to partition a set of n elements is given by the n -th Bell Number, denoted by B_n .
 - For $n=1$, $B_1 = 1$ $\{\{1\}\}$
 - For $n=2$, $B_2 = 2$ $\{\{1,2\}\}, \{\{1\},\{2\}\}$
 - For $n=3$, $B_3 = 5$ (This was our question!)
 - For $n=4$, $B_4 = 15$

Remembering the first few Bell numbers can save you a lot of time in an exam!

Practice Suggestions

- PYQ:** Look for JEE Main questions asking for the number of equivalence relations or the number of onto functions (which also relates to partitions via Stirling Numbers). This is a recurring concept.
- NCERT:** Re-read the "Types of Relations" section in the Class 12 NCERT Chapter 1 (Relations and Functions) to solidify your understanding of Reflexive, Symmetric, and Transitive properties.

3. ****Challenge Yourself:**** Try to list all 15 partitions for the set $\{1, 2, 3, 4\}$. This will give you a great feel for how the combinations work. You nailed it! Keep connecting concepts like this, and you'll do great. Keep up the hard work

Ask your queries

give easiest explanation of the given question

Ask

Imagine you have a group of three friends: Alice, Bob, and Carol. The question is asking how many different ways you can group these three friends together based on a special rule. This rule is called an "equivalence relation." Think of it like this: if Alice is friends with Bob, and Bob is friends with Carol, then Alice *must* also be friends with Carol. They are in the same "group" or "equivalence class." So, the question is asking: how many different ways can you divide Alice, Bob, and Carol into groups, where everyone in a group is considered "equivalent" to everyone else in that same group? The possible ways are: 1. ****Everyone is separate:**** Alice is in her own group, Bob is in his own group, and Carol is in her own group. (3 groups) 2. ****Two are together, one is separate:**** For example, Alice and Bob are in one group, and Carol is in her own group. You could also have Alice and Carol together, or Bob and Carol together. (3 ways to choose which two are together) 3. ****All three are together:**** Alice, Bob, and Carol are all in one big group. (1 group) Adding these up, there are $3 + 3 + 1 = 7$ different ways to group them according to this special rule. The "me @7" and the numbers in parentheses (3, 5, 4) seem to be related to potential answers or a multiple-choice selection, and the actual answer seems to be ****7****.

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